

cells of many animal species, where they appear to help reestablish appropriate methylation patterns in the genome during gamete formation.

➔ **Mastering Biology Interview with Haifan Lin:**
Discovering piwi-interacting RNAs and
connections to testicular cancer



Researchers have also found a relatively large number of **long noncoding RNAs (lncRNAs)**, ranging from 200 to hundreds of thousands of nucleotides in length, that are expressed at significant levels in specific cell types at particular times. The functional significance of these lncRNAs has been debated, but in 2017 a large international research consortium published an atlas of almost 28,000 such RNAs; their analysis supported the idea that almost 20,000 were functional and that some were associated with specific diseases. One lncRNA, long known to be functional, is responsible for X chromosome inactivation, which prevents expression of genes located on one of the X chromosomes in most female mammals (see Figure 15.8). In this case, lncRNAs—transcripts of the *XIST* gene located on the chromosome to be inactivated—bind back to and coat that chromosome. This binding leads to condensation of the entire chromosome into heterochromatin.

The examples just described involve chromatin remodeling in large regions of the chromosome. Because chromatin structure affects transcription and thus gene expression, RNA-based regulation of chromatin structure is sure to play an important role in gene regulation. Additionally, some experimental evidence supports the idea of an alternate role for lncRNAs in which they can act as a scaffold, bringing DNA, proteins, and other RNAs together into complexes. These associations may act either to condense chromatin or, in some cases, to help bring the enhancer of a gene together with mediator proteins and the gene's promoter, activating gene expression in a more direct fashion.

Given the extensive functions of noncoding RNAs, it is not surprising that many of the ncRNAs characterized thus far play important roles in embryonic development—the topic we turn to in the next section. Embryonic development is perhaps the ultimate example of precisely regulated gene expression.

CONCEPT CHECK 18.3

1. Compare miRNAs and siRNAs, including their functions.
2. **WHAT IF?** Suppose the mRNA being degraded in Figure 18.14 coded for a protein that promotes cell division in a multicellular organism. What would happen if a mutation disabled the gene for the miRNA that triggers this degradation?
3. **MAKE CONNECTIONS** Inactivation of one of the X chromosomes in female mammals involves a noncoding RNA (*XIST*). Suggest a model for how *XIST* RNA initiates Barr body formation (see Concept 15.2).

For suggested answers, see Appendix A.

CONCEPT 18.4

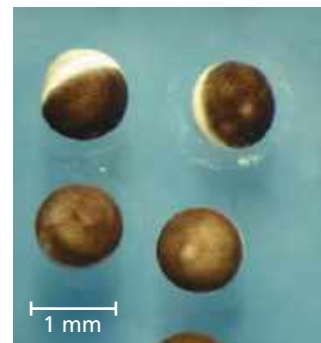
A program of differential gene expression leads to the different cell types in a multicellular organism

In the embryonic development of multicellular organisms, a fertilized egg (a zygote) gives rise to cells of many different types, each with a different structure and corresponding function. Typically, cells are organized into tissues, tissues into organs, organs into organ systems, and organ systems into the whole organism. Thus, any developmental program must produce cells of different types that form higher-level structures arranged in a particular way in three dimensions. The processes that occur during animal development are detailed in Chapter 47; in this chapter, we focus on the program of regulation of gene expression that orchestrates development, using a few animal species as examples.

A Genetic Program for Embryonic Development

The photos in **Figure 18.16** illustrate the dramatic difference between a frog zygote (fertilized egg) and the tadpole it becomes. This remarkable transformation results from three interrelated processes: cell division, cell differentiation, and morphogenesis. Through a succession of mitotic cell divisions, the zygote gives rise to a large number of cells. Cell division alone, however, would merely produce a great ball of identical cells, nothing like a tadpole. During embryonic development, cells not only increase in number, but also undergo cell **differentiation**, the process by which cells become specialized in structure and function. Moreover, the different kinds of cells are not

▼ **Figure 18.16 From fertilized egg to animal: What a difference four days makes.** It takes just four days for cell division, differentiation, and morphogenesis to transform each of the fertilized frog eggs shown in (a) into a tadpole like the one in (b).



(a) Fertilized eggs of a frog



(b) Newly hatched tadpole